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0 3 Printer ink is a mixture of chemicals.

0 3 . 1 What is the name given to a mixture that has been designed as a useful product?

[1 mark]

Tick (✓) **one** box.

Formula ☐

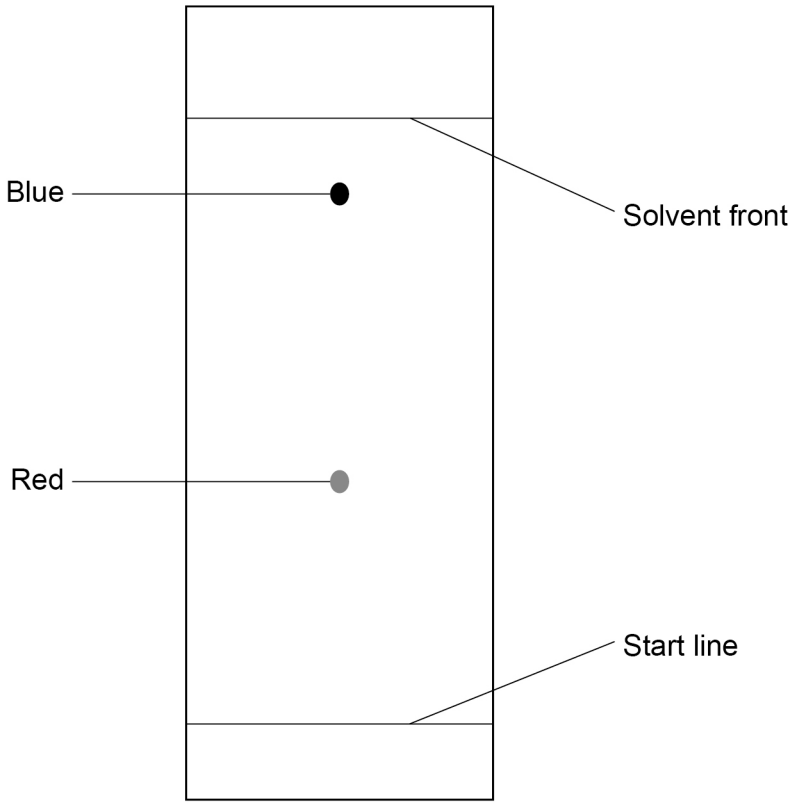
Formulation ☐

Fraction ☐

A student used chromatography to investigate the colours in a printer ink.

Figure 2 shows the chromatogram.

Figure 2



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0 3 . 2 The student used a ruler for the start line.

What would the student have used to draw the start line?

Give **one** reason for your answer.

[2 marks]

_____

Reason _____

_____

0 3 . 3 Determine the R_f value of the **red** colour.

Use the equation:

$$R_f = \frac{\text{distance moved by colour}}{\text{distance moved by solvent}}$$

[4 marks]

Distance moved by red colour _____

Distance moved by solvent _____

_____

_____

_____

R_f = _____

Question 3 continues on the next page

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0 3 . 4 **Table 2** shows the R_f values of four blue colours.

Table 2

Blue colour	**R_f**
Cerulean	0.40
Cobalt	0.15
Prussian	0.88
Ultramarine	0.68

The student determined that the R_f value of the blue colour in the printer ink was 0.86

Suggest which blue colour was used in the printer ink.

Give **one** reason for your answer.

[2 marks]

Blue colour _____

Reason _____

_____

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